radiation treatment options for patients presenting with extensive high risk or recurrent head and neck disease. We present a series of these patients to assess the feasibility of the use of SBRT in both settings. SBRT is used to increase the biologic equivalent dose received by the tumor volume while avoiding excess dose to critical organs at risk (OAR) located within close proximity to the gross tumor volume (GTV). SBRT was given upfront as lone modality, as a boost after sub-radical IMRT dosing, or alone after recurrence.

Materials/Methods: A prospectively collected database was retrospectively analyzed. Between March 2012 and January 2013, eight patients were treated with SBRT to head and neck tumors. An internally reviewed care path for SBRT in the upfront and recurrent setting was used to define treatment parameters including dose limitations to OAR. GTV was defined as residual tumor or high risk postoperative bed, clinical target volume (CTV) was defined as GTV with a margin of 0-1mm for in situ disease or 2mm for surgical bed, and no margin was given for planning target volume (PTV). Endpoints assessed were dose delivered to critical organs at risk, local disease control, and RT related morbidity.

Results: Age at diagnosis ranged from 17 to 95 years of age. 75% of the patients were male. Pathologies included squamous cell carcinoma (n = 4), adenoid cystic (n = 1), atypical pleomorphic adenoma (n = 1), spindle cell (n = 1), and paraganglioma (n = 1). In 5 patients IMRT (range, 44-59.4 Gy) and SBRT (18 Gy in 3 fractions or 25 Gy in 5 fractions) were used upfront with radical intent. One Patient received SBRT alone (25 Gy in 5 fractions) for curative treatment. Two patients were treated with SBRT (25 Gy in 5 fractions) in the recurrent setting. Median follow-up was 4 months after the completion of SBRT. No acute grade III toxicities or above were recorded. One patient developed a grade II neurosensory event in the form of trigeminal neuralgia 3 months after SBRT in the setting of radical surgery and IMRT as well. Radiographically, local tumor control was accomplished in 100% of patients to date.

Conclusions: SBRT allows for the achievement of radical equivalent doses to tumor volume while sparing adjacent critical structures both in the setting of high risk disease or re-irradiation for recurrence in the head and neck. To date, our data suggests SBRT may be used with minimal adverse events and good tumor control yet more information is needed to further define the role of SBRT in head and neck cancers.

Author Disclosure: M. Goldberg: None. A. Whitton: None. J. Wright: None. I. Hodson: None. S. Sagar: None. D. Kim: None. T. Chow: None. S. Gauld: None. J. Greenspoon: None.

2594

Evidence of Efficacy From Radiation Therapy Following Surgery for Oral Cavity Squamous Cell Carcinoma

M.P. Herman,[1] R. Dagan,[2] R.J. Amdur,[1] C.G. Morris,[2] J.W. Werning,[3] M. Vaysberg,[3] and W.M. Mendenhall[1]; [1]*Department of Radiation Oncology, College of Medicine, University of Florida, Gainesville, FL,* [2]*University of Florida Proton Therapy Institute, Jacksonville, FL,* [3]*Department of Otolaryngology, College of Medicine, University of Florida, Gainesville, FL*

Purpose/Objective(s): To evaluate the efficacy of postoperative radiation therapy for the treatment of oral cavity squamous cell carcinoma (SCC) by comparing the outcomes of high-risk subgroups.

Materials/Methods: We reviewed the medical records of 139 consecutive patients who met the following criteria to assess their outcomes: (1) SCC of the oral cavity treated with gross total resection and postoperative radiation therapy +/- chemotherapy. (2) At least one high-risk finding in the pathology report from the resection specimen: positive margin (52%), close (0.1-5 mm) margin (27%), or extra-capsular nodal extension (45%). Most patients (86%) were treated with conventional fields and 14% were treated with intensity-modulated radiation therapy (IMRT). The median dose to the high-risk planning target volume (PTV) was 72 Gy (range, 60-74.8 Gy). Chemotherapy was administered concurrently in 18% of patients (14% cisplatin, 3% cetuximab, and 1% carboplatin).

Poster Viewing Abstract 2594; Table

	5-year control (%)
Local control*	Local control
Negative margin (>5 mm)	73
Close margin (0.1-5 mm)	83
Positive margin (carcinoma in situ or tumor at ink)	64
Neck control*	Neck control
Node negative	100
Node positive with no extracapsular extension	88
Node positive with extracapsular extension	86
Local-regional control†	Local-regional control
Positive or close margin and extracapsular extension	32
All other patients	71

* P value exceeded .05 for all comparisons.
† $P = .0001$.

Results: The median time of follow-up was 2.3 years. Local-regional control, freedom from distant metastases, cause-specific survival, and overall survival at 5 years were 64%, 85%, 51%, and 36%, respectively. Major outcome analyses are summarized as follows: (Tables).

Conclusions: Local-regional recurrence was the predominant mode of treatment failure. These results are indirect evidence that postoperative radiation therapy improves the cure rate in many patients with high-risk SCC of the oral cavity. We observed similar outcomes for subgroups that are known to have different prognoses following surgery alone. Our findings are likely the result of the aggressive radiation therapy that patients received in this series. Patients with a combination of close/positive margins and extracapsular extension are at a significantly high risk of treatment failure.

Author Disclosure: M.P. Herman: None. R. Dagan: None. R.J. Amdur: None. C.G. Morris: None. J.W. Werning: None. M. Vaysberg: None. W.M. Mendenhall: None.

2595

Proton Therapy in Uveal Melanomas: How Relevant Is It to Spare the Macula When the Optic Disk Is Involved?

J.O. Thariat, F. Lanza, M. Peyrichon, C. Barnel, C. Mosci, J. Caujolle, G. Angellier, M. Cabannes, W. Sauerwein, and H. Joel; *Cyclotron-Centre Lacassagne, Nice, France*

Purpose/Objective(s): The optic nerve has a serial architecture. High-dose radiation-induced damage to the optic nerve by proton therapy usually becomes functionally significant after 2 to 5 years. The macula and fovea are composed of millions of photoreceptors, structures involved in the fine central vision (diurnal and colored vision/cones, nocturnal/sticks). There is hardly any data on the behavior of the macula and fovea in terms of dose response patterns and kinetics of vision loss in irradiated parapapillary melanomas, as nerve damage leads to full vision loss. However, empirical data suggest that macular damage occurs before 2 years. In some parapapillary tumors where preservation of the optic disk is not possible, the macula might be spared at the price of a more complex treatment plan (fixation, bolus, and filter) for functional benefit. To investigate whether patients with optic disk irradiated at full dose are more likely to retain some visual acuity (VA) within 2 years of irradiation when the macula is spared (null hypothesis) vs not (alternative hypothesis).

Materials/Methods: Proton therapy planning was performed using the eye model. Kinetics of VA were assessed at baseline, 6, 12, 18, 24 and 60 months in T1-3 N0M0 parapapillary melanoma patients treated in 1996-2006 with baseline VA $\geq$ 4 and with follow-up longer than 5 years with functional data of VA. Cases had 100% optic disk and 0% macula included in the 50% and 90% isodoses and controls had 100% optic disk and macula included in the 50% and 90% isodoses. Cases were compared to controls using chi-square and T tests.